Design and Implementation of High-Speed, Low Power Carry Look-Ahead Adder Using Hybrid GDI Logic and Transmission Gates

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***Abstract*—This paper presents a high-speed, low-power 4-bit Carry Look-Ahead Adder using a hybrid approach by integrating Gate Diffusion Input (GDI) logic and Transmission Gates (TG). The proposed design employs GDI logic to minimize transistor count and reduce power consumption in non-critical paths and ensure full voltage swing with low delay for critical carry propagation paths using TG. The hybrid CLA as compared to the traditional CMOS and fully GDI-based implementations shows major improvement in terms of power, delay, and area. It was synthesized and simulated using Cadence Virtuoso software at 90nm technology and the design resulted in the power reduction by 44.9%, delay by 38%, and area by 30% while maintaining functional reliability for different frequencies and loaded conditions. This novel hybrid CLA design is very appropriate for energy efficient and performance-driven applications in modern integrated circuits.**

Keywords*—GDI, transmission gates (TG), carry look ahead adders (CLA), ripple carry adder (RCA).

I. Introduction

Adders are critical components in digital systems, widely used in processors, signal processing units, and ASICs. Among the various adder architectures, the Carry Look-Ahead Adder (CLA) is preferred for high-performance applications because it reduces carry propagation delay [1][2]. Unlike ripple-carry adders (RCAs), CLAs compute carry signals in parallel using propagate (Pi) and generate (Gi) logic, achieving significantly higher speeds. Traditional CLA designs rely on static CMOS logic for robustness and noise immunity but suffer from high power consumption, transistor count, and area overhead, making them unsuitable for energy constrained systems [3]. With the growing demand for compact, energy-efficient, and high-speed designs, hybrid logic styles have emerged as a solution. Gate Diffusion Input (GDI) logic reduces transistor count, power consumption, and layout complexity while providing efficient logic implementation. GDI alone would not guarantee strong performance in carry propagation paths where TG logic is necessary to get full swing in voltage, minimum delay, and large driving strength. This contribution presents a hybrid CLA design combining GDI style for non-critical paths, and TG style for the critical carry propagation paths, so that it exploits advantages of both styles to better optimize power, delay and area, yet still achieving robust performances [4]. The simulated results at 90 nm technology node using Cadence Virtuoso show improvements that are notable in Average power, Power-Delay Product (PDP) and Propagation delay. This validates the designed circuit's suitability for VLSI modern low-power systems and high-speed systems which may be used in a portable device, processor, and other real time application systems [5].

Many hybrid approaches have been pursued in an effort to optimize the performance of CLA. Traditional static CMOS logic has been widely deployed due to its strength, but power and area overheads tend to be very large. To alleviate these limitations, Pass Transistor Logic (PTL) and Transmission Gates (TG) were implemented to reduce transistor count in non-critical paths and preserve low delay in critical carry propagation paths [6]. In addition, GDI logic has been noticed for its potential to implement fewer transistors for gates like AND and XOR, which save power and area. Significant improvements in PDP and layout efficiency have been achieved in the previous work, and further optimization is possible [3].

In this paper, a new 4-bit Carry Look-Ahead Adder (CLA) is proposed with a hybrid design consisting of Gate Diffusion Input (GDI) logic for non-critical paths and Transmission Gate (TG) logic for critical carry paths. The hybrid combination minimizes transistor count, power consumption, and delay while preserving full voltage swing and signal integrity. In contrast to previous designs our approach optimizes power, speed, and area economically and is well suited for low-power, high-speed VLSI systems.

II. Literature Review

Adders are central building blocks in digital processors and ALUs, where their performance has a significant impact on speed and efficiency. Carry Look-Ahead Adders (CLAs) are favored for their high-speed carry computation. Different logic styles have been used over the years to enhance their power, speed, and area performance.

In [1], CLAs with CMOS technology were incorporated in array multipliers using QSD techniques to lower switching activity and raise speed. Though this helped deal with power, the higher quaternary logic complexity discourages VLSI implementation. In [2], GDI-based CLA implementations over 90 nm, 130 nm, and 180 nm nodes illustrated power and area savings. Still, GDI-only solutions usually experience poor voltage swing on key paths. Work in [3] compared several adders at leading-edge nodes and presented a high speed CLA with smaller PDP and area. Still, KSA continued to outperform in terms of speed, and design complexity was

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high. In [4], an 18-transistor hybrid full adder with Cadence Virtuoso exhibited less power and PDP, but the work targeted isolated adder cells only and not the integration of CLA.

All currently available designs lack either high signal integrity or scalability. To remedy this, the proposed hybrid CLA incorporates GDI logic on non-critical paths and Transmission Gates (TG) on critical paths with complete voltage swing, minimal delay, and lower power. The above balance renders it ideal for VLSI low-power, high performance systems.

III. Methodology

A. Ripple Carry Adder Hybrid Design

The Ripple Carry Adder (RCA), is perhaps the simplest adder architecture: it is constructed by cascading Full Adders. Each stage computes sum and carry, where carry-out of one stage serves as the carry-in for the next [8]. Although RCA is area efficient and easy to design, the sequential carry propagation introduces significant delays, especially in larger bit-width designs. The limitation of this is overcome by the Carry Look-Ahead Adder, which uses propagate (Pi) and generate (Gi) terms to calculate carry signals in parallel. RCAs are widely used as building blocks in hybrid CLA designs, thus balancing simplicity with performance [8][9]. The Ripple Carry Adder is one of the highest-propagation delay adders. The carry of every single bit propagates sequentially, in this case. The CLA avoids this by computing the signals in parallel, significantly lowering the delay for high-speed applications.

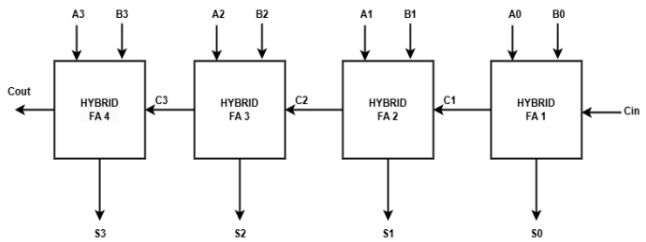


Fig. 1. Hybrid RCA Block diagram

B. Carry Look Ahead Adder Design

The Carry Look-Ahead Adder, or CLA, is an approach to the delay problems of Ripple Carry Adders by calculating carry signals in parallel. Unlike waiting for the carry to propagate sequentially, CLA uses carry generate terms and carry propagate terms to predict carry values at each stage [6].

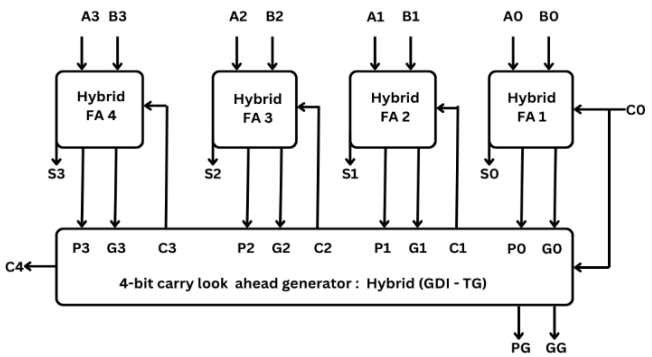


Fig. 2. Hybrid Full Adder Block diagram

The carry-out of any bit position is computed directly as $C(i+1) = Gi + Pi \cdot Ci$, where $Gi = Ai \cdot Bi$ and $Pi = Ai \oplus Bi$. This parallel computation reduces propagation delay significantly; hence, CLA is excellent for high-speed and real-time applications.

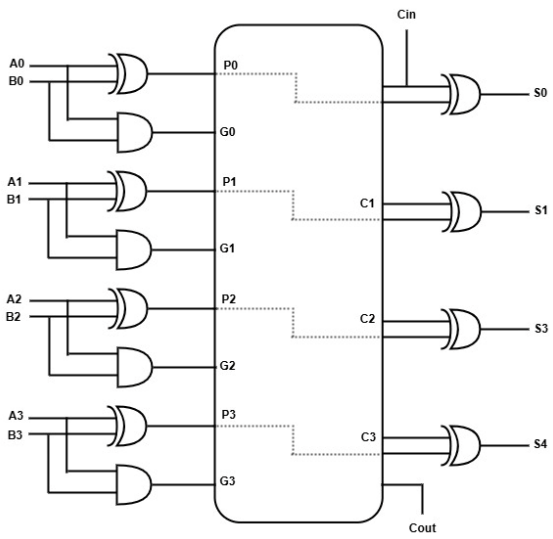


Fig. 3. CLA Gate level diagram

The carry generator circuit is a very important constituent of a CLA. Using generate and propagate signals helps determine whether to generate a carry at a bit position. This way, the generate signals help create a carry directly from the input bits and propagate lets an incoming carry be passed through. The circuit simultaneously pre-computes all the carry values for all bits [6][7]. This means that there is no sequential propagation in the addition process; hence, it makes the operation faster. Therefore, this CLA achieves a drastic reduction in delay compared to other conventional adders. This is highly efficient for high-performance arithmetic units used in modern processors.

The generate signal and propagate signal are defined as follows:

$$Gi = Ai \cdot Bi \tag{1}$$

$$Pi = Ai \oplus Bi \tag{2}$$

The behavior of the generate and propagate signals can be interpreted as follows. For the generate signal Gi, if Ai and Bi are both 1, then Gi = 1, i.e., a carry is produced in that bit position. Otherwise, Gi = 0, and no carry is produced from that position. Similarly, for the propagate signal Pi, it is equal to 1 if Ai or Bi equals 1 but not both, meaning that an incoming carry will be propagated through. If Pi = 0, then the carry-in will not influence the sum at that bit position.

The sum output and carry output equations are given by:

$$Si = Pi \oplus Ci \tag{3}$$

$$Ci+1 = Gi + Pi \cdot Ci \tag{4}$$

The Boolean equations for a 4-bit CLA are derived as:

$$C1 = G0 + P0 \cdot Cin \tag{5}$$

$$C2 = G1 + P1 \cdot C1$$
$$= G1 + P1 \cdot G0 + P1 \cdot P0 \cdot Cin \tag{6}$$

$$C3 = G2 + P2 \cdot C2$$
$$= G2 + P2 \cdot G1 + P2 \cdot P1 \cdot G0 + P2 \cdot P1 \cdot P0 \cdot Cin \tag{7}$$

$$C4 = G3 + P3 \cdot C3 = G3 + P3 \cdot G2 + P3 \cdot P2 \cdot G1 + P3 \cdot P2 \cdot P1 \cdot G0 + P3 \cdot P2 \cdot P1 \cdot Cin \tag{8}$$

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TABLE I. TRUTH TABLE FOR FULL ADDER

A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

TABLE II. TRUTH TABLE FOR A 4-BIT CLA

A	B	C	C i+1	Status
0	0	0	0	
0	0	1	0	No Carry
0	1	0	0	Generate
0	1	1	1	
1	0	0	0	No Carry
1	0	1	1	Propagate
1	1	0	1	Carry
1	1	1	1	Generate

This table demonstrates how Carry Look-Ahead Adder does the computation of carry bits with maximum speed performance, much higher than the simple binary adding method. A, B, and C are the inputs to the system. In the system, A and B are the summands and C is the carry from previous stages. C i+1 is the carry out and "STATUS" tells whether a carry is produced or passed. For the truth table, three possible cases are covered [8]. First case: If A and B are 0, nothing is carried out. Second case: If A ≠ B and C = 1, carry the result from the previous step. This CLA always has a carry in A and B with a value of 1, significantly accelerating addition compared to ripple-carry addition by reducing delay in propagating carries. This CLA is an inevitable component in high-speed digital circuits [4][5].

C. Hybrid GDI and TG Design

Using GDI as well as Transmission Gate is two techniques, through which Carry Look-Ahead Adder can be synthesized with differing benefits. CLA design through GDI consumes low powers and reduces transistor counts based on the usage of the logic used in propagate and generate circuit that minimizes transistors, increases the area efficacy and reduces the power dissipating. In this method voltage swing is restricted along with low margins for noise; thus is unreliable for signal integrity [9][10]. TG-based CLA uses the use of transmission gates for carry propagation; full voltage swing, less signal degradation, and a minimal amount of resistance enhance speed. Despite its advantages, the use of TG logic also needs more transistor numbers which imply increased power consumption as well as increased complexity during the layout. To resolve both problems, a hybrid design that merges GDI for propagate and generate logic with TG-based CLA for carry propagation exists [11].

This approach benefits from the power efficiency of GDI and signal integrity of TG, making it highly optimized in terms of speed, power, and area. The hybrid CLA reduces propagation delays and minimizes the dissipation of power with reliable signal quality, which is why it is so efficient for VLSI systems [12]. This design is best suited to high-speed, low-power applications, including processors and digital signal processing units where both performance and energy efficiency are concerned. The hybrid CLA is perfectly suitable for high-speed, low-power VLSI application as it encompasses both GDI and TG techniques. It can be applied extensively in modern processors where the arithmetic operations are needed to be fast in ALUs and floating-point units. It is also very well-suited for the DSP systems which demand fast addition operations that consume minimum energy, required in the tasks of filtering, convolution, and signal analysis [2][3]. This hybrid CLA can be adopted for low-power applications such as mobile phones, IoT, or wearable devices. Saving energy is paramount for such a group of devices, and speed and power efficiency, particularly with its performance in high-computation computing systems, can ensure adoption [13].

D. Hybrid Full Adder Design

The Full Adder is the building block of designing complex adders, like the Carry Look-Ahead Adder. In this work, a hybrid Full Adder has been designed using GDI and TG logic with an optimization for power, delay, and area. Here, GDI has been applied to XOR and AND gates for the generation of the propagate and generate terms [7]. This reduces the transistor count and power consumption in non-critical paths to a large extent. In critical paths, TGs are used so that full voltage swing, high driving capability, and low propagation delay can be achieved [9][10]. This hybrid approach leverages on the strengths of GDI for compact, low power logic and TG for robust, high-speed performance to make the Full Adder efficient and amenable for VLSI implementations [9]. This proposed Full Adder is simulated using Cadence Virtuoso at the 90 nm technology node, making it the initial block of the CLA design.

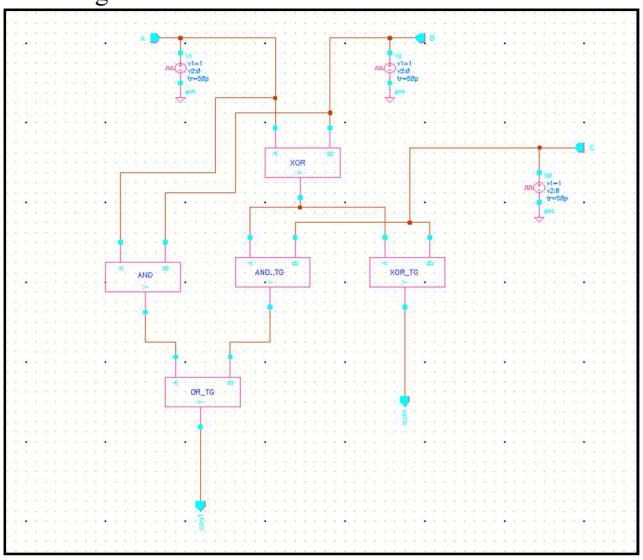
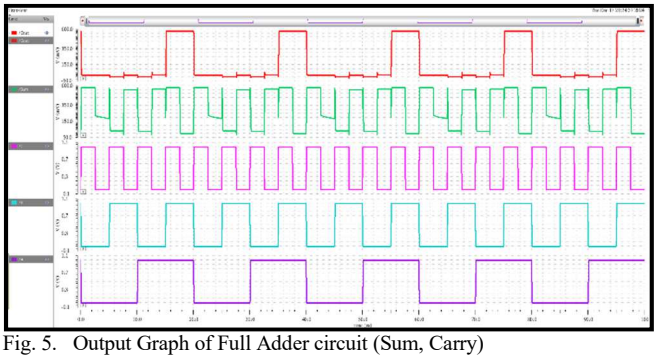


Fig. 4. Full Adder circuit design in Cadence


Fig. 5. Output Graph of Full Adder circuit (Sum, Carry)

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The circuit diagram here and the simulation waveform testify the design of full adder through hybrid model which involves integration of Gate Diffusion Input (GDI) and Transmission Gate (TG) logic. Care was taken in designing this to optimize power, delay and area for low power applications in VLSI. The TG logic is utilized on noncritical paths to reduce transistor counts for power consumption reduction while critical paths use GDI for full swing voltage with fast signal transmission. Majorly, the following are utilized: AND (GDI), XOR (TG), AND-TG and OR-TG, whose objective is to provide a productive solution for computing Sum and Cout of the adder [6].

The simulation waveform validates the functional behavior of the full adder. The inputs A, B, and C are binary signals that depend on time; the produced outputs Sum and Cout should correspond to the expected one in logic. Sum is determined by XOR operation, that is (A $\oplus$ B $\oplus$ C); in the similar way, Cout can be found according to logic, that is (AB + BC + AC). The waveform clearly shows proper transitions and alignment with input changes. This hybrid model significantly reduces power consumption and area due to GDI implementation while TG logic ensures signal reliability and speed [11]. The approach properly balances power, delay, and performance to make it a good foundation for designing high-performance adders like the Carry Look-Ahead Adder (CLA).

E. Hybrid Carry Look-Ahead Adder Design

The Hybrid Carry Look-Ahead Adder (CLA) design combines strengths of Gate Diffusion Input (GDI) logic with Transmission Gate (TG) logic to achieve optimized power, delay, and area performance. In this design, the GDI logic is applied to non-critical paths in an effort to reduce transistor counts and minimize power consumption; TG logic is implemented in the critical carry path so that high-speed operation with a full voltage swing can be preserved. The hybrid CLA overcomes the delay limitations of traditional Ripple Carry Adders (RCA) by parallelizing the carry computation through propagate and generate signals [3][4]. This approach provides superior speed efficiency, low power dissipation, and reduced area, which makes it ideal for high performance VLSI systems [5].

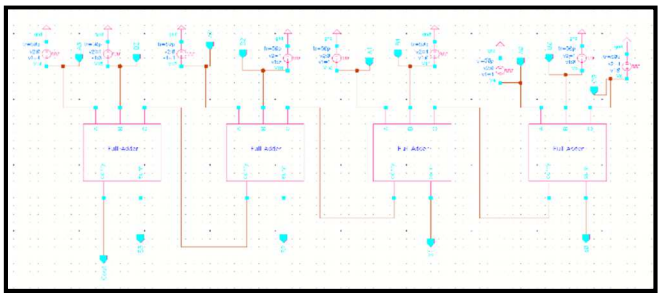


Fig. 6. CLA circuit design in Cadence

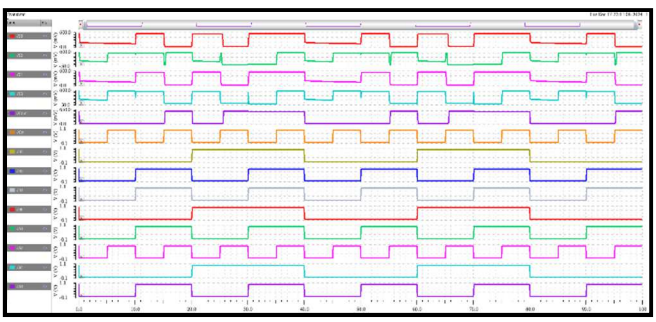


Fig. 7. Output Graph of CLA circuit (Sum, Carry)

The above circuit diagram, along with the simulation waveform, presents a Carry Look-Ahead Adder (CLA), based on a previously designed full adder that utilizes a hybrid Gate Diffusion Input (GDI) and Transmission Gate (TG) logic model. This hybrid approach optimizes the performance of the adder by using GDI logic in non-critical paths to minimize power consumption and transistor count while using TG logic in critical carry propagation paths to ensure high-speed signal transmission and full voltage swing. In the simulation, inputs to the CLA, A, B, and Cin change with time, and the outputs Sum and Carry are generated as expected from the parallel carry computation in the CLA [14][15]. The Sum output is computed using XOR operations, while the Carry Out is derived through the propagate and generate logic, thus improving speed over the sequential carry propagation of Ripple Carry Adders (RCA).

IV. Result And Comparative Analysis

The proposed hybrid CLA shows a substantial improvement over the traditional one, as it can be inferred from the comparison. The average power consumption is decreased by nearly half, i.e., from 16.44 μW to 9.05 μW, with an improvement of 44.94%. This optimization arises from the efficient use of GDI logic in conjunction with TG logic, which has drastically reduced the power dissipation in the non-critical paths. The delay is reduced to 0.093 ns from 0.150 ns, showing a 38.00% improvement. This improvement indicates the proposed design can be used for high-speed applications since parallel carry generation and propagation are made possible with minimal signal degradation. A vital parameter, the Power Delay Product (PDP), which is a measure of energy efficiency, also improves significantly. The proposed hybrid CLA achieves a PDP of 0.841 fJ compared to 2.730 fJ in the conventional design, which reflects a reduction of 68.19%. This is ideal for energy-efficient portable and embedded systems. Additionally, the number of transistors is reduced from 80 to 56, which results in an improvement of 30.00%, thereby reducing chip area and contributing towards lower power consumption, hence scalable for large-scale integration [16].

The hybrid CLA's superior parameters—lower power, delay, and transistor count—reveal the superiority of this hybrid CLA over traditional designs [17]. Balanced trade-off between power, speed, and area achieved through integration of GDI and TG logic makes it highly suitable for modern VLSI systems requiring optimized performance. The design overcomes the limitations of traditional approaches in this respect and paves the way for further innovations in arithmetic logic circuits that address the challenges of power density and performance in advanced semiconductor technologies [17].

The hybrid CLA proposed here manifests balanced enhancements to all key performance parameters, illustrating its efficacy. The rigorous approach to design, optimized partitioning of logic, and successful simulation confirms the completion of this task. This architecture presents an engaging solution for forthcoming arithmetic units when energy efficiency, small area, and speed remain top priorities of contemporary VLSI systems.

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TABLE III. RESULT AND COMPARISON TABLE

Parameters	Result		Improvement
	Existing	*Proposed*	
Average Power (μW)	16.44	9.05	44.94%
Delay (ns)	0.150	0.093	38.00%
PDP (fJ)	2.730	0.841	68.19%
No. of Transistors	80	56	30.00%

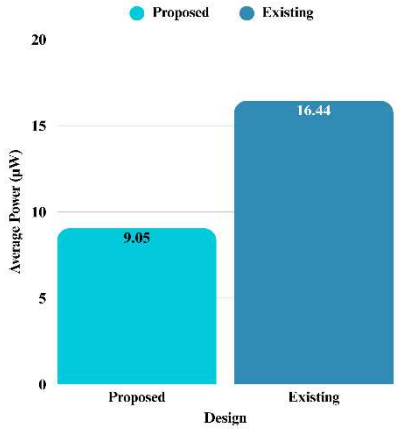


Fig. 8. Comparison of average power

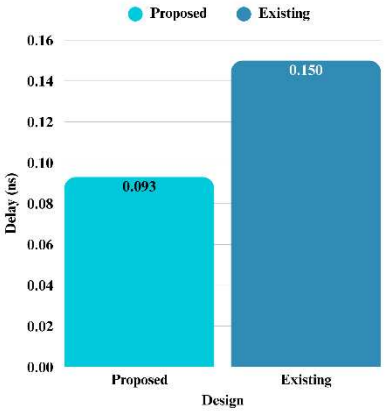


Fig. 9. Comparison of delay

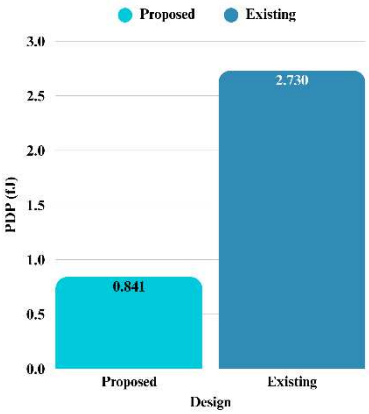


Fig. 10. Comparison of power delay product (PDP)

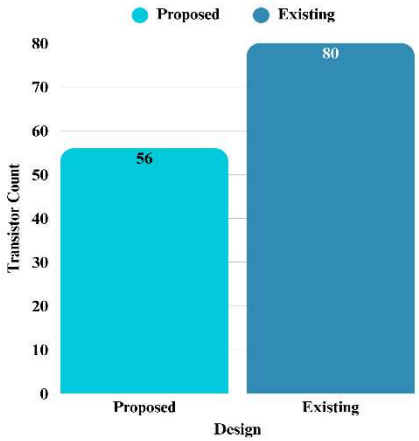


Fig. 11. Comparison of transistor count

V. CONCLUSION

In this paper, the authors proved a low-power high-speed 4-bit CLA through a novel hybrid approach using GDI logic and TG. In comparison with standard CMOS and full GDI designs, this design reduces the power consumption by 44.9%, delay by 38%, and the number of transistors by 30%. This makes the hybrid CLA very suitable for energy-efficient and performance-driven applications in modern integrated circuits. This work is new in combining GDI and TG logic to achieve an optimal balance between power, speed, and area. While the non-critical paths minimize transistor count and GDI power, the critical paths ensure full voltage swing and faster signal propagation through TG. Such efficient hybridization is very useful to realize the high reliability and scalability of the proposed CLA for applications such as portable devices and embedded systems. In the future, it is necessary to extend this hybrid CLA design to higher-bit adders, that is, 8-bit and 16-bit adders to verify its scalability and performance. Further power efficiency can be achieved by using advanced techniques such as clock gating and body biasing. The adaptation of design to FinFET and nanoscale technologies can also provide solutions to the problems of power density and leakage, thus making it quite relevant for next-generation VLSI systems. In this regard, the hybrid approach offers an exciting platform for future innovations in arithmetic logic.

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